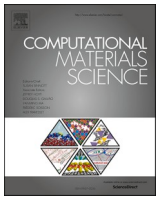
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Full Length Article



# Scaling law-informed machine learning for predicting thermal and electrical properties of polymers: A physics-based approach

Han Xu [a], Xuexian Yu [a], Jun Liu [b], Xiang Gao [a,*]

[a] The State Key Laboratory of Chemical Engineering, College of Chemical and Biological Engineering, Zhejiang University, 38 Zheda Road, Hangzhou 310027, China
[b] Zhejiang Yanyi New Energy Technology Co., Ltd. 29 Xinggong 8th Road, Jiangshan Economic Development Zone, Quzhou, Zhejiang Province, China, 324123

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ABSTRACT

Polymer materials hold great promise for a wide range of applications due to their unique structures and properties. Using machine learning to predict their properties is a highly promising approach. However, the limited number of polymer databases hinders the application of machine learning methods. In this work, we choose to introduce physical prior knowledge at the output layer of the model by fitting the scaling law relationships satisfied by polymer to predict their properties. This approach of introducing physical prior knowledge at the output layer does not conflict with other methods of incorporating prior knowledge, such as molecular encoding and model pre-training, indicating its good portability and potential for application in more polymer property prediction models. We design the model structure from a perspective that aligns with physicochemical intuition. We first validate the model's effectiveness and portability on a small dataset containing 1,070 data points and finally use this method to successfully predict the electrical conductivity of polymer electrolyte materials with an $R^2$ value of approximately 0.8. This modeling approach is expected to open up new avenues for polymer property prediction.

## 1. Introduction

Polymer materials are ubiquitous, found abundantly in nature and widely employed in human manufacturing and everyday life. Their diverse and unique physical properties have led to applications spanning numerous industries, including automotive, energy, and healthcare [1–6]. This ubiquity underscores the critical need to design polymers with tailored structures to achieve specific target properties[7–9]. However, predicting polymer properties from first principles remains a formidable challenge. The sensitivity of polymers to temperature and entropy, coupled with their large molecular sizes, often renders ab initio methods computationally intractable[1,10]. In contrast, machine learning has emerged as a powerful tool for predicting polymer properties. By leveraging statistical methods to discern complex patterns within large datasets, machine learning offers a data-driven approach to polymer design and discovery. This approach, increasingly termed "polymer informatics"[11–13] has already yielded significant advances. Polymer informatics aims to streamline the design process and reduce the reliance on time-consuming and resource-intensive experimental methods, ultimately enabling the development of novel polymer materials with enhanced performance for a wide range of applications. Notably, machine learning has shown promise in developing novel dielectric materials[14,15], solid-state polymer electrolytes (SPEs) [16–19], thermally conductive polymers[20], gas separation membranes[21–23] and more.

Despite the promise of machine learning in polymer science, data scarcity poses a significant bottleneck to its widespread adoption. Researchers have addressed this by pre-training models on large-scale datasets, including supervised transfer learning[24,25] and unsupervised pre-training tasks[26–28]. Another approach involves optimizing datasets with polymer-specific encodings, including BigSMILES[29,30], PolyGrammar[31], PUFPs[32], graph-based encodings[33–37], scaled-composite encodings[38], natural language-based encodings[39] and so on. Comparative studies have evaluated the effectiveness of these different methods[40]. Additionally, active learning methods have been used to gather "in-house" experimental data[41], enabling model training on more complex datasets like non-homopolymer materials [42,43] and polymer-protein hybrids[44]. These approaches introduce structural constraints that enhance performance on smaller datasets: encoding methods incorporate physical and chemical prior knowledge

* Corresponding author.
*E-mail address:* gaox@zju.edu.cn (X. Gao).

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at the input stage, while pre-training and active learning embed self-learned prior knowledge within the model architecture. Although effective, encoding methods increase data processing complexity and create interoperability issues between databases, whereas pre-training and active learning require significant resources.

Physics-Informed Machine Learning (PIML) models have been applied to various fields[45–48]. PIML aims to constrain the search space of the model by embedding physical knowledge into it, resulting in better accuracy and generalization with limited and imperfect data[49]. Compared to PINNs[47,50–53] that simulate partial differential equations (PDE) in the loss function, a more direct implementation of PIML outputs parameters of established physical equations rather than directly predicting properties—an approach sometimes referred to as chemistry-informed machine learning[18]. For example, models may output parameters of the Arrhenius equation to calculate polymer electrolyte conductivity[18], parameters of the Flory-Huggins theory to predict phase separation in polymer condensed states[54], or parameters of the Nernst equation to compute electrode material potentials [55]. Unlike methods discussed previously, PIML introduces physical and chemical constraints at the output side, offering a simpler approach that just requires modifying the neural network's output layer. Importantly, PIML exhibits inherent compatibility with various data encoding schemes, model pre-training tasks, and other model constraints applied at the input or model architecture level. This integration of complementary approaches presents opportunities for further enhancing predictive accuracy by combining the strengths of different constraint-base.

In this work, we propose a method for predicting polymer properties by incorporating scaling laws into the output layer of neural network models. The concept that properties of soft materials like polymers follow scaling laws was first introduced by Nobel laureate de Gennes in 1979[56]. Polymer chains can be represented as "coiled springs" that can be stretched and compressed, wherein the statistical behavior of each chain section is shown to be identical to that of the whole chain – a manifestation of self-similarity across different length scales. This remarkable property, which is known as self-similarity, indicates that consistent mathematical descriptions can be obtained whether a small section or the entire chain is examined. Such scale invariance enables the prediction of polymer properties using simple power-law equations. By incorporating these power-law relationships into neural networks, we can create models that naturally respect the physical principles governing polymer behavior, leading to more accurate predictions even with limited data. Dobrynin et al. [57] used scaling law informed CNNs to predict polymer solution properties, but their study was limited to a single model and property. Recognizing that scaling laws can characterize numerous physical relationships in polymers, we designed a scaling law informed machine learning model (called as SLIMNet), from a physics-informed perspective. We first validated the model's effectiveness, portability (compatibility with other algorithms), and performance on a small-size datasets. Further, we confirmed SLIMNet's efficacy by predicting the ionic conductivity of polymer solid electrolytes, establishing a novel approach for machine learning-assisted polymer property prediction.

## 2. Methods

### 2.1. Concept of scaling laws in polymer materials

Polymer materials demonstrate characteristic properties that follow well-defined scaling laws. These laws arise from the inherent self-similar structure of polymer chains, where statistical patterns remain consistent across different length scales. Under external stimuli such as thermal or mechanical stress, polymers respond in mathematically predictable ways. A notable example is the relationship between molecular weight and viscosity, which follows a power law relationship – a fundamental behavior that reflects the underlying molecular interactions. This theoretical framework, first established by de Gennes in 1979, provides a systematic approach to connect molecular-level structure with macroscopic properties. The self-similar characteristics of polymer chains align naturally with the mathematical principles of ***fractals*** – structures that preserve their patterns independent of scale.

A ***fractal*** is a geometric concept used to describe certain types of structures. As illustrated in Fig. 1a with the Sierpinski triangle, these patterns exhibit specific symmetry often referred to as self-similarity and scaling invariance. When scaled up or down, the image retains its shape. The simplest mathematical representation of this self-similarity is the scaling law equation, typically expressed as a product of a pre-factor and a power of a physical quantity, for example:

$$y = k \bullet x^{\alpha}$$

The Sierpinski triangle demonstrates how a geometric pattern remains invariant under scaling transformations, which can be directly analogous to how polymer chain segments retain their conformational characteristics across different length scales (Fig. 1b). Polymer chains are characterized by their inherent flexibility, which permits their segmentation into statistically independent units called Kuhn segments. When described by Kuhn segments, polymer chains can be mathematically represented as 'Gaussian segments' (Fig. 1b), wherein the spatial distribution between any segments is governed by a Gaussian function. This statistical behavior is found to be scale-independent – the same mathematical principles apply across all length scales, leading to the emergence of scaling laws that govern polymer properties. For example, in dilute polymer solutions, the root mean square end-to-end distance $h$ of the polymer chain scales with the molecular weight $N$ as:

$$h = b \bullet N^{\nu}$$

where $b$ is the length of a Kuhn segment, determined by the polymer's chemical structure, and NN is proportional to the molecular weight. The exponent $\nu$ is universal for different polymers: 0.3 in poor solvents, 0.5 in $\theta$ solvents, and 0.588 in good solvents[58]. Similar relationships exist for other properties like zero-shear viscosity and diffusion coefficients in polymer melts. These scaling relationships have been verified by numerous experimental results.

These assumptions form the core of scaling theory of polymers and are the theoretical foundation for the design of SLIMNet.

### 2.2. Assumptions of SLIMNet

To effectively utilize scaling laws for predicting polymer properties within SLIMNet, we establish two key assumptions:

**Firstly**, we recognize that real polymers deviate from the ideal conditions assumed by scaling laws, which presume perfectly flexible chains and negligible inter-chain interactions. To address this, we introduce the concepts of "ordered phase" and "disordered phase." The disordered phase embodies the ideal scaling law conditions, allowing

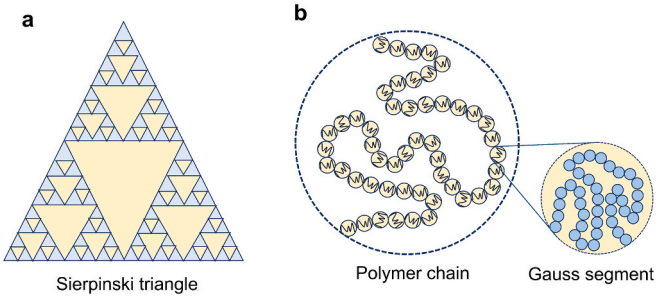



**Fig. 1.** **a** Sierpinski triangle, a fractal pattern that remains invariant under arbitrary scaling. **b** From a conformational distribution perspective, polymer chains also exhibit fractal properties, as the conformations of both the entire polymer chain and the polymer molecules within the Gaussian chain segments follow a Gaussian distribution.

direct property calculation using scaling laws. Conversely, the ordered phase encompasses the possible excluded volume effect, interchain friction, and chain rigidity that may exist in polymer materials. The properties corresponding to the ordered phase cannot be directly predicted using the scaling law because they do not meet the basic assumptions of the scaling law. It is supposed to understand that these phases are not physically distinct regions within a real polymer but rather conceptual constructs for our model. In SLIMNet, we predict the property contributions from both phases separately. The disordered phase contribution is determined directly from scaling laws. For the ordered phase, we utilize order parameters, which capture the degree of order within the polymer structure, to predict its property contribution. The final predicted property is the direct sum of these two contributions. We deliberately avoid introducing weighting factors for these contributions, assuming that the neural network can learn the appropriate balance from the training data, as this balance reflects the intrinsic nature of the specific polymer.

**Secondly,** for polymer chains, the range of scales at which scaling laws apply extends from the length of a Gaussian segment to the entire chain length. Hence, we assume the neural network can derive the properties of Gaussian segments from the input monomer chemical structure and use these properties in the scaling law equations to predict the properties of the entire chain. According to our first assumption, when calculating properties for the disordered phase, the input to the scaling law is the information of Gaussian segments, thereby adhering to the basic assumptions of scaling laws. The implementation of these assumptions in SLIMNet's design will be detailed in the next section.

### 2.3. Framework of SLIMNet

Based on the above assumptions, we designed the model structure of SLIMNet (Fig. 2). The complete SLIMNet consists of two components: the base model and SLIMNet module.

#### 2.3.1. Base model (Blue Area)

This component focuses on generating an accurate monomer representation vector, denoted as $V_{monomer}$. Theoretically, any suitable small-molecule property prediction model can serve as the base model. It takes a monomer molecule as input and, through feature extraction steps specific to the chosen base model, outputs a predicted monomer property (denoted as $\hat{\phi}_{monomer}$). This prediction is supervised by the ground truth monomer property ($\phi_{monomer}$) during training. The variable from the last hidden layer of the base model is extracted as the monomer representation vector $V_{monomer}$.

#### 2.3.2. Slimnet module (Yellow Area)

This module leverages the monomer representation vector and scaling laws to predict polymer properties. The $V_{monomer}$ vector is fed into the SLIMNet module, where it's combined with polymer chain parameters to produce a polymer representation vector $V_{polymer}$. To clarify, this $V_{polymer}$ effectively represents the property of a Gaussian segment. We use "polymer property" here for simplicity and to avoid potential confusion, as there isn't a well-defined physical interpretation for the property of a Gaussian segment. Importantly, subsequent calculations involve neural network layers, allowing for interpretation of these properties at the polymer chain level.

Following the model assumptions, $V_{polymer}$ is used to predict the properties corresponding to the ordered phase $\phi_{ordered}$ and the disordered phase $\phi_{disordered}$ :

- $\phi_{disordered}$ is predicted using scaling laws. Based on the scaling law assumption, $\alpha$ is derived from $V_{monomer}$, and $\beta$ and $\gamma$ are derived from $V_{polymer}$ through the neural network, with the final calculation as $\phi_{disordered} = \alpha \odot \beta^{\gamma}$.
- $\phi_{ordered}$ is obtained by directly concatenating the order parameter of the polymer molecule and processing it through a neural network.

The final polymer property is calculated as:

$$\hat{\phi}_{polymer} = \phi_{disordered} + \phi_{disordered}$$

Both $\hat{\phi}_{monomer}$ and $\hat{\phi}_{polymer}$ are included in the loss function, functioning as a multi-task network. The model's loss function is defined as:

$$Loss = \lambda * MSELoss(\hat{\phi}_{polymer}, \phi_{polymer}) + MSELoss(\hat{\phi}_{monomer}, \phi_{monomer}) + \phi^2_{ordered}$$

Where *MSELoss* is defined as:

$$MSELoss(\hat{\mathbf{y}}, \mathbf{y}) = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$$

$N$ represents the number of samples, and $\lambda$ is a tunable hyper-

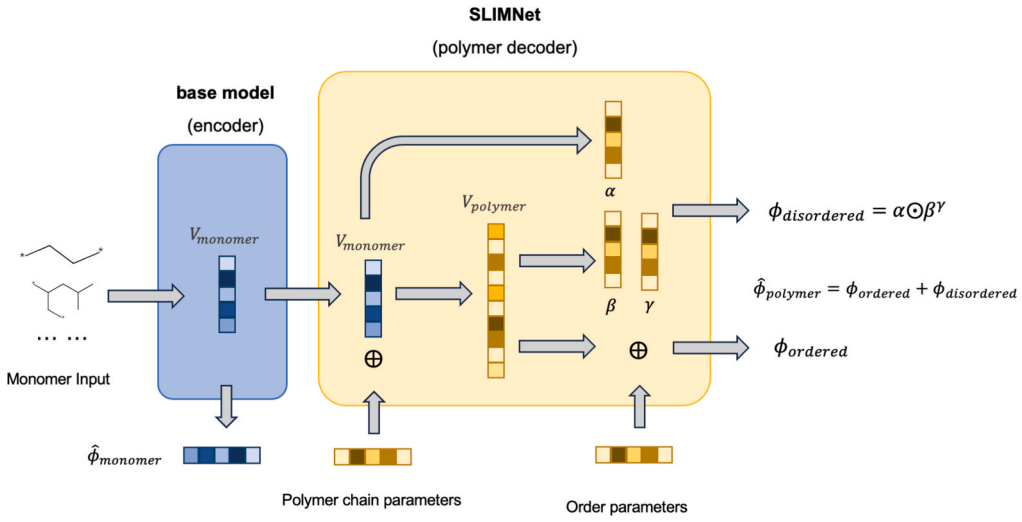



**Fig. 2. Model structure of SLIMNet.** The blue area represents the base model, used to calculate the properties of monomers, which theoretically can be replaced by any small molecule property prediction model. Its input is the molecule of a monomer, and the output is the property corresponding to the monomer molecule. The yellow area represents SLIMNet module, which combines scaling laws to make polymer property prediction. Its input consists of the last hidden layer variable $V_{monomer}$ from the base model, polymer chain parameters and order parameters, while the output includes the property corresponding to the disordered phase calculated by scaling laws and the property corresponding to the ordered phase calculated by combining the order parameter.

parameter depending on the training task. As $\lambda$ decreases, the model assigns more training weight to the prediction of monomer properties, aiming for a more accurate monomer representation. The term $\phi^2_{ordered}$ acts as a regularization term, reducing the emphasis on ordered phase properties during loss minimization. This encourages the model to rely more on the property prediction branch containing prior knowledge, improving performance on small datasets, the square operation ensures that the penalty term is positive, as will be demonstrated in the subsequent experiments.

### 2.4. Data acquisition and processing

#### 2.4.1. PI1M

PI1M dataset is used for self-supervised pretraining to get Mol2Vec [59] embedding. PI1M was built by Ma et al. by training a generative model on polymer data collected from the PolyInfo database[60,61]. It includes ~ 1 million polymer molecules, covering similar chemical space as PolyInfo but filling gaps where PolyInfo data is sparse. The pretraining process for this dataset follows the methods and relevant code provided in the original article[59].

#### 2.4.2. QM9

QM9 dataset[62,63] is used for supervised pretraining (transfer learning) in SchNet[64]. QM9 provides quantum chemical properties (at DFT level) for a relevant, consistent, and comprehensive chemical space of small organic molecules. QM9 dataset contains 134 K stable small organic molecules with geometric, energetic, electronic, and thermodynamic properties.

#### 2.4.3. Small-size dataset

Small-size dataset was obtained from the publicly available dataset in RadonPy[65,66], contains 1070 polymer molecules. The specific description of the dataset can be found in the official documentation. The purpose of selecting this dataset is to test the performance of SLIMNet in a process closer to the actual material design. Polymeric materials have a wide range of applications. Taking dielectric packaging materials as an example, good polymeric materials do not depend on a single property. They must have a high dielectric constant, and the thermal expansion coefficient of the polymeric material must be close to that of the substrate to ensure good thermal conductivity and mechanical properties. At the same time, the actual screening range is relatively small, and the collected dataset will not be very large. Therefore, in this experiment, we used 1070 groups of smaller data and selected thermal conductivity, thermal diffusivity, linear thermal expansion coefficient, and static dielectric constant as the prediction targets. In the base model part, the p-SMILES of the repeating unit form (marked with * at the repeating link) were selected as the input[67], and HOMO, LUMO, dipole moment and polarizability were selected as the supervision of $\widehat{\phi}_{monomer}$ (the output of base model), and the molecular weight, degree of polymerization, and density of the polymer were selected as the chain sequence parameters, and nematic order was selected as the order parameter.

#### 2.4.4. SPE dataset

SPE (Solid-state Polymer Electrolyte) dataset used comprised 10,114 samples of polymer electrolyte molecular data obtained from publicly available datasets published in peer-reviewed articles[18,68]. Each sample contains six pieces of information: the polymer electrolyte molecular structure and its corresponding molecular weight information, the solvent molecular structure, and its corresponding concentration information, as well as temperature and ionic conductivity. Combined with the description in the original paper, a complete solid polymer electrolyte contains the synergistic effect of polymer molecules and corresponding solvents. To facilitate comparison with the results in the original paper, the input for the base model section includes the repeating unit structure of the polymer and the structure of the solvent molecule. The polymer chain structure information included the molecular weight of the polymer, and the order parameters of the polymer were determined to include the concentration of solvent molecules and the temperature. It is worth noting that due to the difficulty in determining the properties of polymer monomers and solvent molecules considered as a whole, and also to be consistent with the original paper's included information to facilitate comparison, in this experiment, $\widehat{\phi}_{monomer}$ will not be calculated.

## 3. Results and discussion

### 3.1. Validation of SLIMNet on small-size dataset

#### 3.1.1. Comparison with traditional algorithm

We first validate the performance of SLIMNet by comparison. Given the relatively small dataset size, theoretically, tree-based models often outperform neural network models in relatively small tabular data[69]. Consequently, we chose Random Forest as a benchmark against SLIMNet, utilizing a standard three-layer neural network as the base model (Comparison of Neural Network can be seen in supplementary materials Fig. S10). Both algorithms were fed with MACCS[70] fingerprints, and all predicted properties were standardized to mitigate the influence of varying scales.

Fig. 3 illustrates the comparative performance, employing Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared ($R^2$) as evaluation metrics. MAE and RMSE gauge the model's prediction accuracy, with lower values indicating better performance. Conversely, $R^2$ assesses the goodness of fit, with values closer to 1 signifying stronger predictive power (more details in Supplementary Materials Part 1). While SLIMNet outperforms Random Forest across all three metrics, indicating superior overall performance, the scatter plots reveal that Random Forest achieves reasonably accurate predictions in several instances. The significantly lower $R^2$ value for Random Forest primarily is attributed to severe prediction errors for a few molecules.

Further analysis of these results reveals that Random Forest's predictive accuracy diminishes for data points deviating significantly from the mean (normalized to 0), as depicted in Fig. 3a. In contrast, SLIMNet exhibits superior fitting for such outliers. Consider the thermal conductivity prediction: Fig. 3a demonstrates that Random Forest struggles to accurately predict three samples with normalized ground truth values around −4, negatively impacting its $R^2$ coefficient. Conversely, SLIMNet, as shown in Fig. 3b, achieves relatively accurate predictions for two of these samples, with only poly(1-(but-2-en-2-yl)-4-iodobenzene) (monomer **C**) exhibiting a less satisfactory result.

This discrepancy can be attributed to the underlying assumptions of SLIMNet. Our model presupposes that most polymer chains exhibit flexibility, enabling behaviors like folding and entanglement. However, in the case of poly(1-(but-2-en-2-yl)-4-iodobenzene) (monomer **C**), the bulky side groups introduce significant steric hindrance and inter-chain friction, hindering segmental motion and causing its properties to deviate from those predicted by scaling laws. This analysis is corroborated by the bulk modulus prediction: poly(hepta-1,5-diene) (monomer **D**), lacking bulky side groups and possessing greater chain flexibility, is predicted more accurately by SLIMNet (Fig. 3b) compared to Random Forest (Fig. 3a). Due to the limited quantity and inadequate visualization characteristics of the RadonPy data, the variations in properties of ordered phase and disordered phase output by SLIMNet will be more intuitively elucidated in the subsequent sections pertaining to the design of polymer electrolytes.

This analysis of extreme value predictions (monomer A, B, C and D) reveals a fundamental advantage of physics-informed models over purely data-driven approaches. While traditional machine learning methods like Random Forest excel at interpolating within the main distribution of training data, they inherently struggle when predicting

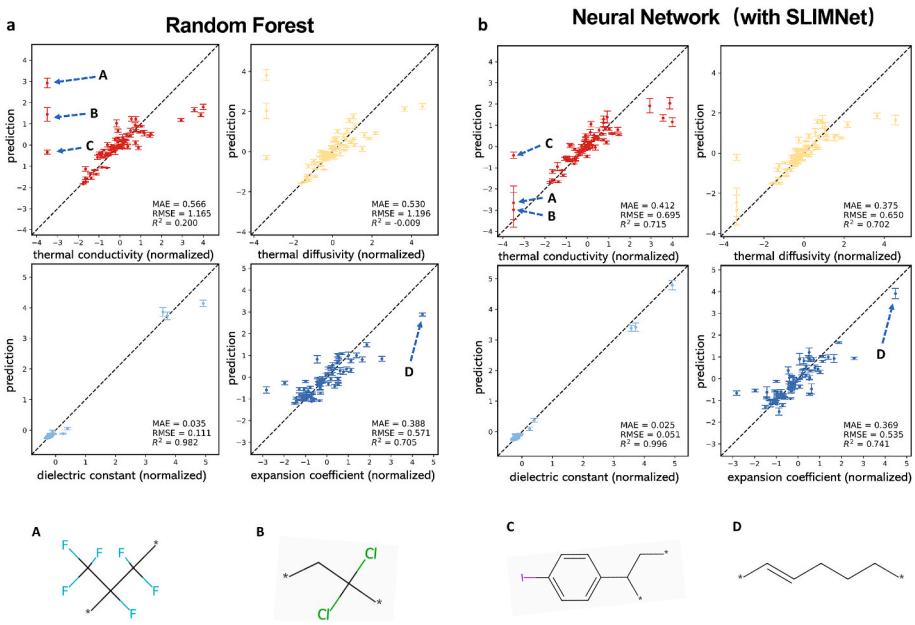



**Fig. 3.** The comparison between the random forest algorithm **a.** and SLIMNet **b.** The x-axis represents the actual property values (ground truth); the y-axis represents the predicted values. The red points represent thermal conductivity, the yellow points represent thermal diffusivity, the light blue points represent static dielectric constant, and the dark blue points represent linear thermal volume expansion.

properties that deviate significantly from the mean. This limitation stems from their reliance on statistical patterns rather than underlying physical principles. In contrast, SLIMNet's architecture, by incorporating scaling laws and physical constraints, provides a natural framework for extrapolating to these extreme cases. The model's success in predicting properties for molecules like poly(1-(but-2-en-2-yl)-4-iodobenzene) and poly(hepta-1,5-diene) demonstrates that it has learned genuine physical relationships rather than merely capturing statistical correlations. This capability is particularly crucial for polymer design, where identifying materials with exceptional properties often requires accurate predictions at the extremes of the property distribution.

### 3.1.2. *Parameter sensitivity and ablation study*

Building on the previous experimental setup, we further investigate the impact of parameter $\lambda$ in the loss function on the model, as well as conduct an ablation study on the regularization term $\phi^2_{ordered}$ (as shown in Fig. 4a and b, respectively). All experiments were conducted using three rounds of random cross-validation.

The purpose of setting parameter $\lambda$ is to balance the model's focus between predicting polymer properties and monomer properties. The experimental results show that by altering various $\lambda$ values, SLIMNet's MAE displayed a smile-shaped curve in response to this parameter. As $\lambda$ increases, the weight of the loss function for the entire polymer chain grows, while the weight for the monomer properties diminishes. This reduction in monomer property training results in an imprecise monomer being used as the unit input for the scaling law, leading to less accuracy in parameter $\alpha$ and a decline in model performance. Conversely, as $\lambda$ decreases, the training emphasis shifts more to the monomer properties at the expense of the polymer chain properties, which directly reduces the overall training accuracy of the model. In this dataset, the adverse impact of inadequate monomer training due to a large $\lambda$ is less severe than that resulting from insufficient training of the polymer chain itself. When $\lambda$ equals 100, the model achieves a lower training error.

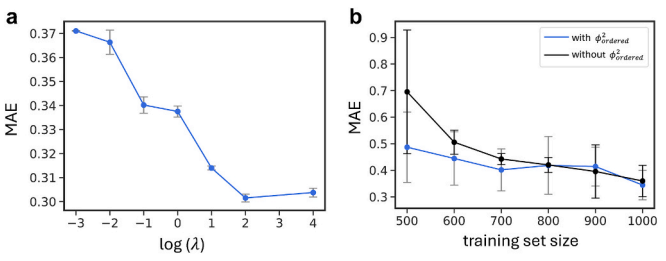



**Fig. 4.** Impact of **a** loss function weight parameter λ and **b** regularization term $\phi^2_{ordered}$ on SLIMNet's performance, measured by Mean Absolute Error (MAE).

The purpose of the regularization term $\phi^2_{ordered}$ is to channel more information through the output layer that incorporates scaling laws, thereby allowing the inclusion of more prior knowledge into the output data. To explore this, we performed an ablation study by reducing the dataset size. The experimental results indicate that as the dataset size decreases, the model's output error increases. Additionally, the error of the model without the regularization term $\phi^2_{ordered}$ increases more rapidly as the dataset reduces. This suggests that the presence of $\phi^2_{ordered}$ directs the data flow through the branch containing the scaling laws, introducing a degree of prior knowledge, which helps maintain model performance even with smaller datasets.

### 3.1.3. *Portability of SLIMNet*

SLIMNet's design incorporates prior scaling law information directly into the output layer. Theoretically, this approach should be compatible with other techniques that inject prior knowledge, such as incorporating domain-specific features during molecular encoding or leveraging pre-trained models. This inherent flexibility, termed "portability", allows SLIMNet to be readily adapted to various property prediction models. We designed a series of experiments to validate this portability hypothesis.

As illustrated in Fig. 5a, we selected three distinct molecular property prediction approaches: 1. **Mol2Vec Fingerprints:** Utilizing self-supervised Mol2Vec embeddings trained on the PI1M dataset as input features (more details in Supplementary materials part 1.3). 2. **Graph Neural Networks:** Leveraging automatically extracted molecular features from the message-passing graph neural network SchNet [71]. 3. **Transfer Learning:** Employing a pre-trained model on the QM9 small molecule database to enhance predictive performance. Detailed descriptions of these approaches and the corresponding datasets are

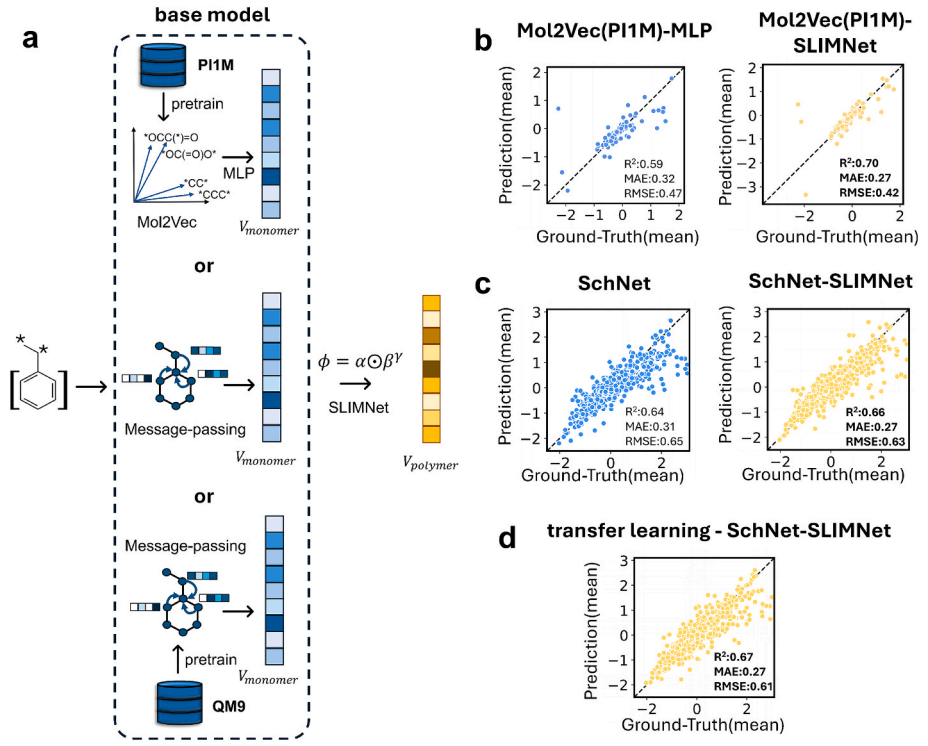



**Fig. 5. Portability of SLIMNet. a.** Using three different methods to obtain monomer representation vectors $V_{monomer}$ as the base model, connected to SLIMNet to acquire polymer representation vectors $V_{polymer}$ based on scaling laws. The specific results of these three methods are depicted in the right column. **b.** Obtaining Mol2Vec encodings via self-supervised pre-training with PI1M, computed using both MLP and SLIMNet. **c.** Direct prediction results using SchNet (with polymer chain structural information as additional input) compared to predictions using the SLIMNet integrated approach (with different test set from experiment in **b**). **d.** Prediction results of the model that further employs transfer learning with LUMO data from the QM9 small molecule database on the SchNet-SLIMNet framework (with same test set from experiment in **c**).

provided in the "Data Acquisition and Processing" section.

For the first approach (Fig. 5b), Mol2Vec fingerprints derived from self-supervised pre-training on PI1M were fed into both a traditional three-layer MLP and SLIMNet. As anticipated, integrating SLIMNet as the output layer consistently yielded superior predictive performance. Similarly, we incorporated SLIMNet into the SchNet (Fig. 5c), utilizing SchNet-derived molecular features as input. Again, SLIMNet demonstrably enhanced the model's performance compared to standalone SchNet. Furthermore, we explored transfer learning (Fig. 5d) by pre-training on the QM9 database using LUMO property data. Subsequently, the model was fine-tuned on our polymer dataset. This strategy led to further performance gains, as evidenced by the experimental results.

It is worth noting that SchNet, by virtue of receiving atomic coordinates as input, inherently captures some conformational features, which aligns with the information introduced by the scaling law. Consequently, the performance improvement from adding SLIMNet to SchNet was less pronounced compared to the other approaches. However, a more nuanced understanding emerges when examining performance across different dataset sizes. To investigate this relationship further, we evaluated SLIMNet-SchNet's performance on progressively smaller datasets (Table 1). The results revealed an important distinction: as the training data diminished, the $R^2$ coefficient of the standard SchNet model deteriorated rapidly, while the SchNet augmented with SLIMNet maintained significantly higher performance. Notably, even with only 300 training samples, the SLIMNet-enhanced model maintained an $R^2$ value above 0.6.

The significance of this $R^2$ threshold warrants careful consideration within the context of materials science. In polymer property prediction, where inherent material complexity and measurement uncertainties play significant roles, $R^2$ values exceeding 0.6 typically indicate strong correlation[72]. This benchmark is particularly meaningful when evaluating models trained on limited datasets, as maintaining such correlations despite data sparsity suggests robust predictive capability. These findings demonstrate that while SchNet's inherent ability to capture conformational features may reduce SLIMNet's comparative advantage with large datasets, the incorporation of scaling laws through SLIMNet provides crucial benefits when working with limited data. By leveraging prior physical knowledge, SLIMNet effectively compensates for data sparsity, enhancing the model's robustness and generalizability in precisely those scenarios where traditional deep learning approaches struggle most.

**Table 1**
Performance of the model with reduced dataset.

| Num of training dataset | Model | MAE | RMSE | $R^2$ |
|---|---|---|---|---|
| 500 | SchNet | 0.332 | 0.706 | 0.574 |
| | **Sch-SLIMNet** | **0.320** | **0.649** | **0.640** |
| 300 | SchNet | 0.350 | 0.725 | 0.511 |
| | **Sch-SLIMNet** | **0.348** | **0.677** | **0.608** |
| 50 | SchNet | 0.516 | 0.939 | 0.246 |
| | **Sch-SLIMNet** | **0.469** | **0.908** | **0.295** |

### 3.2. *Prediction of ionic conductivity of polymer electrolytes*

After verifying the effectiveness of SLIMNet, we extended our validation to a larger dataset comprising polymer electrolytes to further investigate the model's performance in predicting properties of polymer materials in a real-world application. The dataset acquisition and preprocessing steps are elaborated in the **Method** section.

To ensure a fair comparison with previous work, we adopted the same Message Passing Neural Network (MPNN) architecture[73] employed in the original study[18] to generate molecular fingerprints. These fingerprints were subsequently projected onto a two-dimensional space using t-Distributed Stochastic Neighbor Embedding (t-SNE) for visualization, as depicted in Fig. 6a. The axes in this plot represent arbitrary dimensions after projection and lack direct physical interpretation. Each data point signifies a unique "electrolyte molecule + solvent molecule" combination. The t-SNE visualization of molecular fingerprints (Fig. 6a) reveals distinct clustering patterns that provide crucial insights into how our model captures structural relationships among different electrolyte systems. The clear separation between clusters demonstrates that the MPNN-derived fingerprints effectively encode both primary chemical features and subtle structural variations. Particularly noteworthy is the organization of systems sharing the same polymer backbone but different anions – this form adjacent but distinct clusters in the representation space, indicating that our model successfully captures the nuanced effects of anion chemistry on the overall system properties. The relative positioning of these clusters exhibits a remarkable correlation with chemical intuition: systems with similar functional groups or structural motifs are consistently positioned in proximity to each other, while those with distinct chemical features maintain clear boundaries in the representation space.

We reproduced the results of the MPNN and ChemArr methods [18,68]. These results are presented in Fig. 6c & d and Table 2, demonstrating consistency with the original findings across ten-fold cross-validation. We benchmarked the performance of SLIMNet using the same MPNN as the base model under identical experimental settings. The results, obtained via ten-fold cross-validation, are illustrated in Fig. 6e and summarized in Table 2. The direct comparison across the three approaches reveals that SLIMNet consistently outperforms both MPNN and ChemArr, achieving a remarkable Mean Absolute Error (MAE) for ionic conductivity of approximately 0.57 orders of magnitude. The comparative analysis of predictive performance across different models (Fig. 6c–e) demonstrates the capabilities of our physics-informed approach. The baseline ChemProp model exhibits systematic limitations, particularly evident in its handling of high-conductivity systems where the complex interplay between polymer structure and ion transport becomes more pronounced. The ChemArr method, while showing improved performance, still struggles to fully capture the subtle variations in conductivity across different structural and environmental conditions. In contrast, SLIMNet demonstrates consistency and accuracy across the entire conductivity range, with improvements in predicting systems at both extremes of the conductivity spectrum. This enhanced performance can be attributed to the model's ability to separate and appropriately weight the contributions from ideal chain behavior and real-world structural effects. The error distribution in SLIMNet's predictions shows a more symmetric pattern centered around the ideal prediction line, indicating a balanced treatment of various physical factors affecting ion conductivity.

**Table 2**
Performance Metrics for Each Model.

| | Mean Absolute Error (log(S/cm)) | Spearman R |
|---|---|---|
| ChemProp | 0.90 + -0.016 | 0.59 + -0.013 |
| ChemArr | 0.79 + -0.030 | 0.64 + -0.023 |
| SLIMNet | **0.57 + -0.033** | **0.78 + -0.027** |

To gain deeper insights into SLIMNet's predictions, we analyzed the behavior of its two output components: $\phi_{disordered}$ and $\phi_{ordered}$, as visualized in Fig. 6b. This figure focuses on two specific input combinations: PEO + LiTPSB and PEO + LiTf. Each bin represents a group of samples, with blue denoting the property contribution from the ordered phase and yellow representing the disordered phase contribution. Within each

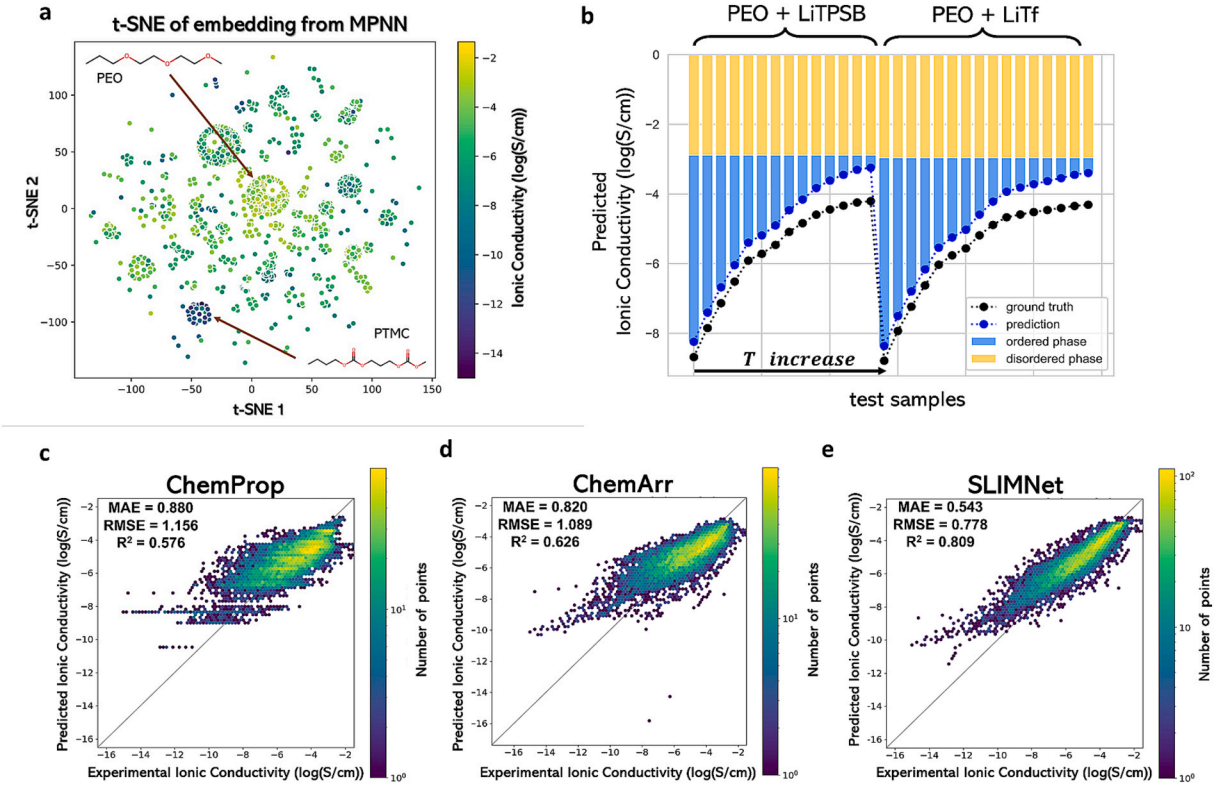



**Fig. 6.** Ionic conductivity prediction results for polymer electrolytes. **a**, t-SNE results for $V_{monomer}$. **b**, The prediction of SLIMNet outputs among different data. For the same 'monomer + electrolyte' combination (test samples), with increasing temperature the property of the disordered phase, determined by the scaling law, is kept constant, while the property of the ordered phase, determined by the order parameter, gradually decreases. This means that SLIMNet operates following physical intuition. **g**, Schematic diagram of the solid polymer electrolyte structure. **c**, *ChemProp* prediction results. **d**, *ChemArr* prediction results. **e**, SLIMNet prediction results.

group, the temperature increases from left to right. Notably, the y-axis utilizes a negative logarithmic scale, signifying an increase in ionic conductivity with rising temperature, aligning with our physical intuition.

From a design perspective, $\phi_{disordered}$ within SLIMNet aims to capture the ideal scaling behavior of the polymer property. In contrast, $\phi_{ordered}$ represents deviations from this ideal behavior, accounting for real-world factors like chain entanglement, steric hindrance, and inter-chain friction. Consequently, for a given input material, the property contribution associated with $\phi_{disordered}$ should primarily depend on intrinsic chemical properties and remain relatively insensitive to external factors such as temperature, as reflected by the consistent blue regions in Fig. 6b. This observation confirms that SLIMNet behaves as expected. Conversely, the $\phi_{ordered}$ component, representing the residual between the idealized scaling prediction and the actual property, exhibits a distinct temperature dependence. As temperature rises, polymer chains gain greater mobility, overcoming conformational energy barriers and exploring a wider range of conformations. This increased chain flexibility contributes to higher ionic conductivity while simultaneously driving the system closer to the idealized behavior captured by $\phi_{disordered}$. This trend manifests as a decrease in $\phi_{ordered}$ with increasing temperature.

The analysis of the SLIMNet outputs for polymer electrolytes provides strong evidence that our model captures genuine physical phenomena rather than mere mathematical fitting. As shown in Fig. 6b, for both PEO + LiTPSB and PEO + LiTf systems, the disordered phase contribution ($\phi_{disordered}$) remains constant across different temperatures, consistent with the fundamental scaling law behavior of ideal polymer chains. This temperature independence aligns with the theoretical understanding that scaling laws describe intrinsic polymer chain statistics that are primarily determined by chemical structure rather than external conditions. In contrast, the ordered phase contribution ($\phi_{ordered}$) systematically decreases with increasing temperature, reflecting the enhanced chain mobility and reduced inter-chain interactions at higher temperatures. This behavior matches our physical intuition as discussed above. The separation of polymer properties into ordered and disordered phase contributions finds strong theoretical support in established polymer physics. The disordered phase aligns with de Gennes' scaling theory, which describes polymer behavior when chains can be treated as ideal random walks. This regime occurs when excluded volume effects and inter-chain interactions are minimal, allowing the statistical nature of chain conformations to dominate. The ordered phase contribution, meanwhile, captures deviations from ideal behavior due to real polymer effects such as chain entanglements, excluded volume interactions, and local conformational restrictions. This physical interpretation is further supported by our observation that $\phi_{ordered}$ decreases with temperature while $\phi_{disordered}$ remains constant – exactly what we would expect if these components truly represented non-ideal and ideal chain behavior, respectively. Thus, rather than being arbitrary fitting parameters, these contributions reflect fundamental physical distinctions in polymer behavior.

## 4. Conclusions

This work reports a neural network model based on scaling laws, SLIMNet, which focuses on modifications to the neural network's output layer and the introduction of prior knowledge. Specifically, the approach allows the neural network to output the undetermined parameters of scaling laws, with results obtained via further computations. The results derived from this method benefit from the inclusion of physicochemical prior knowledge, thereby yielding better outcomes.

Initially, we designed experiments on a smaller dataset of only 1070 molecules to demonstrate the role of each component in SLIMNet's design and its effectiveness on smaller datasets. Furthermore, we validated SLIMNet's transferability. This included the use of self-supervised model pre-training to obtain molecular fingerprints, message-passing graph neural networks to acquire molecular fingerprints, and supervised transfer learning pre-training techniques. In all cases, SLIMNet was proven to further enhance model performance. This endows SLIMNet with the capability to integrate into the latest algorithms at any time, enhancing its performance or adapting it even better to polymer property prediction. Subsequently, we demonstrated SLIMNet's performance on a polymer electrolyte dataset, revealing that its results surpass current leading models. Moreover, the physical interpretability of the ionic conductivity properties of polymer electrolytes allowed us to further verify the rationality of the output properties.

However, the impact of different order parameters on the model remains to be fully explored in SLIMNet's design. This limitation is due to the interpretability and richness of the dataset. We look forward to future efforts in seeking more suitable databases to realize this vision and hope to further investigate the influence of different order parameters on property predictions. Greater levels of physical interpretation could guide material design.

The approach of embedding physical prior knowledge in the output layer may be a fundamental strategy worthy of more attention in materials design models. As large-scale pre-trained language models develop, their basic structure follows the principle of "pre-training large models" + "downstream tasks." This represents a modular philosophy in model design, avoiding the resource waste associated with repeatedly training model bases for different tasks. In the context of materials design, we face a similar situation where numerous types of materials exist, but it is challenging to unify them under a single model. If we can use a method like SLIMNet that introduces physicochemical knowledge constraints in the output layer for other types of materials, we could similarly design a base model specifically for predicting properties of structural units of materials. This could then link to different material design networks, i.e., different downstream tasks. Such an approach could potentially open up a new era in machine learning-based materials design.

## Author contributions

H.-X. designed and carried out the implementation and experiments under the guidance of X.-G. and J.-L. H.-X. wrote the manuscript under the revised and help by X.-G. and X.X.-Y. All authors discussed the results and reviewed the manuscript.

## Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. Additional data related to this paper may be requested from the authors.

## Code availability

https://github.com/huxelyXu/SLIMNet.

## CRediT authorship contribution statement

**Han Xu:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Xuexian Yu:** Writing – review & editing. **Jun Liu:** Supervision. **Xiang Gao:** Writing – review & editing, Project administration, Methodology, Funding acquisition, Conceptualization.

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## Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.commatsci.2025.113887.

## Data availability

Data will be made available on request.

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